

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Canon
Firmware Version: WUX450ST: 01.020000
WUX500: 02.020000
Model(s): WUX450ST, WUX500, WUX500ST

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.06.0002-b002	1.4.1

Version History

Module Version	Date	Notes
1_0_1_0	12/18/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='WUX450ST')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='WUX450ST')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 33336, Model='WUX450ST')</code>

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	Value
AspectRatio	'Auto'	'4:3'	'16:9'
	'16:10'	'Zoom'	'Real'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command	Value		
AutoImage	None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command	Value	Value	Value
Input	'HDMI'	'Digital PC'	'Analog PC-1'
	'Analog PC-2'	'Component'	'LAN'
	'USB'	'HDBaseT'	
# Input example InterfaceName.Set('Input', 'HDMI')			
Command	Value		
Keypad	'0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '0')			
Command	Value	Value	
LampMode	'Full Power'	'Eco'	
# LampMode example InterfaceName.Set('LampMode', 'Full Power')			
Command	Value	Value	Value
MenuNavigation	'Menu'	'Up'	'Down'
	'Left'	'Right'	'Ok'

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Revision: 12/18/2017

	'Exit'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')	
Command Power	Value 'On' Value 'Off'
# Power example InterfaceName.Set('Power', 'On')	
Command VideoMute	Value 'On' Value 'Off'
# VideoMute example InterfaceName.Set('VideoMute', 'On')	
Command Volume	Value 0 to 20 in steps of 1
# Volume example InterfaceName.Set('Volume', 20)	

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AspectRatio	'Auto'	'4:3'	'16:9'
	'16:10'	'Zoom'	'Real'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
DeviceStatus	'No Error'	'Temperature Error'	'Lamp Error'
	'Lamp Cover Error'	'Cooling Fan Error'	'Power Supply Error'
	'Air Filter Error'		
# DeviceStatus examples InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command	Value	Value	
Freeze	'On'	'Off'	

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<pre># Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)</pre>			
Command Input	Value 'HDMI' 'Analog PC-2' 'USB'	Value 'Digital PC' 'Component' 'HDBaseT'	Value 'Analog PC-1' 'LAN'
<pre># Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command LampMode	Value 'Full Power'	Value 'Eco'	
<pre># LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)</pre>			
Command LampUsage	Value '0 to 539' '1620 to 2159' '2850 to 2999'	Value '540 to 1079' '2160 to 2699' '3000+'	Value '1080 to 1619' '2700 to 2849'
<pre># LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command Power	Value 'On' 'Cooling Down'	Value 'Off'	Value 'Warming Up'
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command SignalStatus	Value 'Signal Not Detected'	Value 'Image Displayed'	Value 'Signal Detection In Progress'
<pre># SignalStatus examples InterfaceName.Update('SignalStatus') Value = InterfaceName.ReadStatus('SignalStatus') InterfaceName.SubscribeStatus('SignalStatus', None, FeedbackHandler)</pre>			
Command VideoMute	Value 'On'	Value 'Off'	
<pre># VideoMute examples InterfaceName.Update('VideoMute')</pre>			

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<pre>Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>	
Command Volume	Value 0 to 20 in steps of 1
<pre># Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>	

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

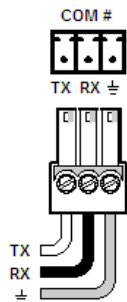
Data Bits: 8

Parity: None

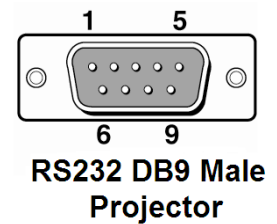
Stop Bits: Two

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	33336
Logon Credentials Supported:	No
Multi-Connection	Yes
Capabilities:	
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 16:10	ASPECT=16:10\x0D
Aspect Ratio 16:9	ASPECT=16:9\x0D
Aspect Ratio 4:3	ASPECT=4:3\x0D
Aspect Ratio Auto	ASPECT=AUTO\x0D
Aspect Ratio Real	ASPECT=TRUE\x0D
Aspect Ratio Zoom	ASPECT=ZOOM\x0D
Audio Mute Off	MUTE=OFF\x0D
Audio Mute On	MUTE=ON\x0D
Auto Image None	AUTOPC\x0D
Freeze Off	FREEZE=OFF\x0D
Freeze On	FREEZE=ON\x0D
Input Analog PC-1	INPUT=A- RGB1\x0D
Input Analog PC-2	INPUT=A- RGB2\x0D
Input Component	INPUT=COMP\x0D
Input Digital PC	INPUT=D- RGB\x0D
Input HDBaseT	INPUT=HDBT\x0D
Input HDMI	INPUT=HDMI\x0D
Input LAN	INPUT=LAN\x0D
Input USB	INPUT=USB\x0D
Keypad 0	RC=NUM_0\x0D
Keypad 9	RC=NUM_9\x0D
Lamp Mode Eco	LAMP=ECO\x0D
Lamp Mode Full Power	LAMP=FULL\x0D
Menu Navigation Down	RC=DOWN\x0D
Menu Navigation Exit	RC=EXIT\x0D
Menu Navigation Left	RC=LEFT\x0D
Menu Navigation Menu	RC=MENU\x0D
Menu Navigation Ok	RC=OK\x0D
Menu Navigation Right	RC=RIGHT\x0D
Menu Navigation Up	RC=UP\x0D
Power Off	POWER=OFF\x0D
Power On	POWER=ON\x0D
Video Mute Off	BLANK=OFF\x0D
Video Mute On	BLANK=ON\x0D
Volume 0	AVOL=0\x0D

Appendix B. Update Commands

Aspect Ratio	GET=ASPECT\x0D
Audio Mute	GET=MUTE\x0D
Device Status	GET=ERR\x0D
Freeze	GET=FREEZE\x0D
Input	GET=INPUT\x0D
Lamp Mode	GET=LAMP\x0D
Lamp Usage	GET=LAMPCOUNTER\x0D
Power	GET=POWER\x0D
Signal Status	GET=SIGNALSTATUS\x0D
Video Mute	GET=BLANK\x0D